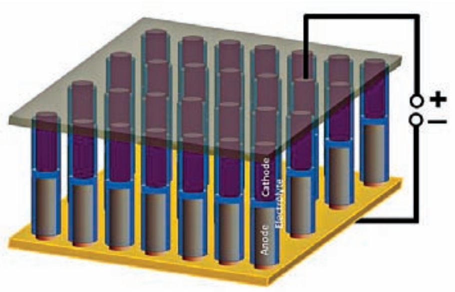
Rice University researchers build nano-sized batteries



An innumerable number of devices run on lithium-ion batteries. Thanks to Rice University researchers, their size is no longer a factor (at least for mobile devices). Prof Pulickel Ajayan and his colleagues claim to have succeeded in building a complete lithium-ion energy storage device in a single nanowire. The researchers say their batteries are as small as devices can get, and hence immensely valuable as rechargeable power sources for the latest nanoelectronics.

The researchers tested two versions of their battery/supercapacitor hybrid. The first version resembles a sandwich comprised of nickel/tin anode, polyethylene oxide (PEO) electrolyte and polyaniline cathode layers. This setup proved that lithium ions could efficiently traverse the anode to the electrolyte, and then make their way back to the supercapacitor-like cathode. The cathode is called a supercapacitor because of its ability to discharge and charge quickly using ions in bulk.

"The idea here is to fabricate nanowire energy storage devices with ultrathin separation between the electrodes," said Arava Leela Mohana Reddy, a research scientist at Rice and co-author of the paper. "This affects the electrochemical behavior of the device. Our devices could be a very useful tool to probe nanoscale phenomenon."

Speaking about the time it will take such batteries to actually hit the streets, Sanketh Gowda, the paper's lead author, said: "There's a lot to be done to optimize the devices in terms of performance. Optimization of the polymer separator and its thickness and an exploration of different electrode systems could lead to improvements." d

AMD reveals pricing and release date of the first 8-core Bulldozer Zambezi FX CPUs

AMD has revealed the approximate retail pricing of its first eight-core Bulldozer processors on a promotional webpage - $300. Part of its upcoming FX-Series, which also includes four and six core processors, the eight-core Bulldozers are codenamed Zambezi, with the first two to be released as the FX-8100 and FX-8150 models.

The AMD Bulldozer FX-8150 is the top-end offering - based on the 32-nm process, it has a 3.6 GHz core clock that can turbo up to 4.2 GHz, all whilst maintaining a TDP of 125 W. At the $300 pricing, AMD puts it roughly at the same price as Intel's current Sandy Bridge Core i7-2600 offering, a quad-core processor. However, AMD expects it to compete directly against Intel's upcoming Sandy Bridge Extreme processors, expected in Q4 2011, and Ivy Bridge processors, expected in Q1 2012.

The first Bulldozer processors are expected to be the Zambezi offerings, and are to be launched in October. Also expected with them are the four-core and six-core FX-4100 and FX-6100 processors. d

Nintendo slashes 3DS prices to boost sales

Nintendo's not doing so well, announcing its first quarterly operating loss, and that it would be slashing prices of the Nintendo 3DS portable gaming console. The move

worries market watchers, who think it will further affect the company's falling stock prices, which dropped over 30% in the past quarter.

The Japanese video gaming giant still intends to sell 16 million of its projected 3DS units, but at 15,000 yen, the maximum suggested retail price in Japan has been cut by nearly 40%. The 3DS will be priced at $169.99 in the U.S., two-thirds its current going rate.

The quarterly operating loss is Nintendo's first recorded since 2003, which was when the company first started declaring quarterly earnings. While Nintendo is not the only Japanese company to be struggling to recover after the Tohuku earthquake, most rival consumer electronics companies, including Sony, have managed to stay above the line in contrast.

The next-generation Wii U, announced for 2012, is Nintendo's greatest hope to bolster its flagging position in the market, and we hope the Japanese giant manages to create an attractive offering in terms of both hardware and software. A lot of developers have already signed up to create titles for it, with the likes of Ninja Gaiden, Assassin's Creed, Darksiders, and Arkham City all scheduled to make an appearance. d